Bijux Pollenomics: scientific evidence

Sources, method correspondence, predictions and model use

Bijux Pollenomics

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Source-bound research briefing

Evidence ladder

1. Archive/source custody and taxon-count mapping.
2. Author-equation and numerical implementation parity.
3. Observed-pollen held-out prediction.
4. LRA reconstruction and forward-pollen prediction.
5. Independent measured-vegetation validation: no admitted pair yet.

Each rung answers a different question; a result at one rung does not imply the next.

Evidence: `reports/README.md`; `reports/needed/README.md`

Four-country source and prediction accounting

Target	Sites	Samples	Assigned/native NISP	Scored/attempted bins
Denmark	1	91	122,998/129,768	24/24
Finland	28	1,797	1,229,019/1,260,281	435/435
Norway	52	3,544	2,018,295/2,286,020	1,084/1,178
Sweden	3	228	221,800/240,966	51/54

Closed 31-target author pollen composition. Excluded/unmapped mass remains outside the score. Zero Nordic samples are admitted to LRA by this study.

Evidence: `reports/science/denmark-finland-norway-author-pollen.md; reports/science/sweden-author-pollen.md`

Source membership is distinct from a scored prediction

- We joined all 12,388 Neotoma source sample IDs to the four-country observed-pollen study: 5,660 selected across 84 sites, 6,218 outside those sites, and 277 without an assigned source country.
- The remaining 233 samples share selected sites but were not selected. The source pollen screen refuses all 233; age and count-candidate reasons overlap for six.
- Six selected Swedish samples also fail the source age screen. Their negative-BP ages map to nominal bin -1, which the frozen Swedish predictor refuses at all three sites; none is scored.

Study membership, source screening and scored site-bins are separate decisions. This is prior-exposed observed-pollen evidence, not LRA or vegetation validation.

Evidence: `reports/science/neotoma-study-membership/README.md`; `reports/science/neotoma-study-membership/report.json`

SEAD is four-country chronology context

Country	Assigned sites	Sites with claims	Chronology claims
Sweden	1,925	741	22,643
Denmark	59	56	1,939
Norway	45	27	468
Finland	40	29	59

The 25,109 rows are chronology claims at 853 distinct sites, not pollen observations or admitted ecological events. A further 103 country decisions require review and 23 remain unassigned. All claim propagation is refused.

Evidence: `reports/science/source-cohorts.md`; `docs/report/source-relations/sead/chronology-country-accounting/report.json`

Three further Swedish lake cores

- Metadata-only selection added Gilltjarnen, Lake Avegol and Flinkasjon outside the original Swedish targets.
- We assigned 231,115 of 238,797 native pollen grains to the 31 author targets; excluded and unmapped counts remain visible.
- All 63 site-bins received three fixed predictions, yielding 189 matched scores with no training-support refusal.
- Equal-site pollen MAE: IDW 0.017334, nearest site 0.017932, training mean 0.020682. Nearest site wins at Lake Avegol.

Three prior-exposed lake groups; direct observed pollen only, with no Nordic LRA or independent vegetation score.

Evidence: `reports/science/sweden-lake-transfer.md`; `docs/report/regions/nordic/swedish-lake-transfer/site-scores.csv`

Swedish training scope on shared pollen targets

- Hold out each Swedish site; train on the other two Swedish sites or those same two plus 81 DK/FI/NO sites.
- On 39 shared site-bins, inverse-distance-squared equal-site pollen MAE is 0.032846 versus 0.023226, favoring four-country training.
- Sweden-only refuses 15 of 54 candidate bins; four-country training predicts all 54.

Three previously exposed sites; most mean gain comes from Abborrtjarnen. No LRA or independent vegetation accuracy follows.

Evidence: `reports/science/sweden-training-scope.md`; `docs/report/regions/nordic/swedish-training-scope/paired-scores.csv`

Swedish source support changes the pollen score

- We froze 1,419 method predictions across 473 supported queries; 13 of 486 site-bin/policy queries refused.
- Adjacent nominal 500-year source bins and a nearest-five cap changed IDW error slightly and differently by site.
- Excluding source centers within 100 km raised matched equal-site IDW MAE from 0.023151 to 0.029171 on 52 bins.

Three exposed Swedish sites; site centers, no age posterior, LRA reconstruction or measured vegetation.

Evidence: `reports/science/sweden-spatiotemporal-sensitivity.md`; `docs/report/regions/nordic/swedish-support-sensitivity/paired-scores.csv`

LegacyAge links two Nordic cores to alternative ages

- LegacyAge S5/S6 join 107/107 ABBO and 71/72 DALMUT depths; the unmatched 4.5 cm DALMUT row stays refused.
- All 178 matched Neotoma point ages lie within the reported 95% ranges; the age models differ.
- ABBO strict support: eight samples, four bins; site-excluded IDW MAE 0.060058 to 0.059762 on 12 paired targets.
- DALMUT strict support: two samples, six frozen 31-target Norway-excluded predictions; 20/22 bins unsupported.

Sparse prior-exposed development evidence. No historical LandClim age, LRA input or independent skill claim follows.

Evidence: `docs/report/source-relations/neotoma/legacyage-chronology/report.json`;

`docs/report/pollen-reconstruction/age-interval-sensitivity/comparison.json`;

`docs/report/regions/nordic/dalmutladdo-author-target-age-interval/report.json`

Solsø: alternative ages restrict Danish pollen scoring

- Primary LegacyAge S5 joins all 92 depths across two Solsø cores; one Neotoma calibrated age stays refused.
- Strict interval containment retains 15/91 comparable samples in eight of 24 bins; both cores contribute.
- The 32 frozen Denmark-excluded predictions yield nearest-site eight-bin MAE 0.032310, versus 0.032104 on complete parent-bin counts.
- Separately, 24 frozen 31-target predictions give nearest-site MAE 0.025673 on author-assigned counts; the axes and denominators differ.

One prior-exposed lake; 16-label and 31-target pollen scores use different denominators. No age reassignment, historical LandClim selection, LRA or vegetation validation follows.

Evidence: `docs/report/countries/denmark/solsoe-source-lineage/legacyage-chronology/report.json`;

`docs/report/countries/denmark/solsoe-source-lineage/legacyage-interval-sensitivity/report.json`;

`docs/report/countries/denmark/solsoe-source-lineage/author-target-age-interval/report.json`

Swedish chronology-block pollen transfer

- Hold out each target site; predict 54 target bins using training pollen exactly 1,000 nominal BP older, excluding the target and adjacent source bins.
- All 54 bins yielded three frozen predictions and matched same-bin scores; zero queries refused.
- Inverse-distance-squared equal-site pollen MAE changed from 0.023647 to 0.024023 on the same targets; nearest-site and training-mean changes were small and in the opposite direction.

Prior-exposed three-site development result; assigned age bins do not account for age-model uncertainty. No Nordic LRA or measured vegetation score follows.

Evidence: `reports/science/sweden-chronology-block.md`; `docs/report/regions/nordic/swedish-chronology-block/paired-scores.csv`

A newly scored Swedish lake on a different pollen axis

- Lake Sämbosjön was absent from earlier targets; all 22 candidate bins received three fixed predictions and scores.
- Inverse-distance-squared equal-bin pollen MAE was 0.022991 on 16 exact source labels, covering 224,291 of 393,822 native pollen counts.
- Aggregate count support was screened before prediction; aggregate scores were viewed before a later protocol annotation correction. Prediction bytes did not change.
- Source age bounds place 58 of 111 samples across a nominal 500-year bin boundary; all 22 scored bins contain a crossing sample.

One-site descriptive development result. The 16-label error is not comparable to the 31-author-target error; no independent confirmation, LRA or measured vegetation score follows.

Evidence: `reports/science/sambosjon-direct-pollen.md`; `docs/report/regions/nordic/sambosjon-direct-pollen/report.json`; `docs/report/regions/nordic/sambosjon-direct-pollen/age-bounds/report.json`

Sämbosjön: restrict the reported age support

- Keep samples whose marginal age interval lies within one nominal 500-year bin: 53 of 111 samples, 21 of 22 bins. Bin 1 is refused.
- With the identical frozen predictions, primary inverse-distance-squared equal-bin MAE is 0.023054 versus 0.022881 for original counts on those same 21 bins.
- Retained native count mass is 185,150; 108,163 counts lie on the 16-label response. Hellinger distance is 0.185741 versus 0.183698 on matched bins.

One exposed core; marginal bounds have no stated confidence level or joint posterior. This sensitivity does not estimate age uncertainty, confirmation or vegetation accuracy.

Evidence: `docs/report/regions/nordic/sambosjon-direct-pollen/interval-support/report.json; interval-support/scores.csv and coverage.csv`

A second Swedish lake: fixed methods differ by metric

- Lilla Gloppsjön site/core was excluded from 84 training sites; 23 bins received 69 predictions before target scoring.
- All 69 scores completed on 16 exact pollen labels: 55,638 of 127,027 native NISP counts lie on the closed response.
- Equal-bin MAE: training mean 0.016272, primary inverse-distance-squared 0.017295, nearest site 0.020000.
- Hellinger: nearest site 0.228724, inverse-distance-squared 0.237576, training mean 0.256392. No method was selected using this target.

One-site descriptive direct-pollen development; the earlier source-validation screen parsed the target partition. Two age-refused samples remain outside scores; no independent LRA or vegetation claim.

Evidence: `reports/science/lilla-gloppsjon-direct-pollen.md`; `docs/report/regions/nordic/lilla-gloppsjon-direct-pollen/scored/report.json`

What reproduces numerically

- Author raw-to-prepared replay: 17 example files, 871 raw sample rows, 15,028 matching prepared cells.
- WHH05: 301 raw grains map to 297 prepared; 31 paired means and SEs agree at 1500 BP.
- Reexecuted author R/native calculations on two grids (9+8 sites); 14,074 supported mean and SE values agree within declared tolerances.
- Archive TW2 differs: total variation 0.092/0.111; two taxa explain 73%/94% of each grid's L1 gap. Historical inputs remain unclosed.
- Archived ISTS and TBS mean arithmetic closes over all 10,500 grid-window rows, but biological classification and SE transformation remain open.

Evidence: `reports/science/publication-correspondence.md`; `reports/science/publication-correspondence/published-window-provenance.md`; `reports/science/publication-correspondence/author-example-reexecution.md`

REVEALS source lineage on one author window

- We ran the 2016 author function and pinned later Petr Kuneš LRA source on the same 225 BP input in four site configurations.
- All 124 paired cover values and 124 paired standard errors agree at retained CSV precision; the full-grid R calls sampled the same 1,000 bootstrap site indices.
- With those indices fixed, our historical native profile matches the tagged R full-grid cover and standard errors within declared tolerances.

Conditional numerical lineage for one window. The R function bodies differ; this does not certify uncertainty correction or Nordic ecological accuracy.

Evidence: `docs/report/method-certification/petrkunes-r-comparison/README.md`; `docs/report/method-certification/petrkunes-r-comparison/summary.csv`; `docs/report/method-certification/petrkunes-r-comparison/native-report.json`

LOVE 2024: paper inputs and unavailable output

- All 30 published fall-speed, PPE and PPE-SD values match the later DISQOVER example by row position; nine of ten taxon labels match exactly.
- The paper's eight simulated ponds, landscape extraction, ring geometry, pollen inputs and 100 random runs are not present in the retained example.
- Figure 2 RMSE and its 100–500 m tendency therefore remain a paper result, not a reproduced Nordic model setting.

Input correspondence is established; the eight-site numerical simulation is blocked on its original inputs.

Evidence: `reports/science/publication-correspondence.md`; `docs/report/method-certification/love-paper-correspondence/README.md`

LOVE optimizers: shared objectives, different searches

- We gave installed RcppDE and Python the same three declared pollen, deposition, background, PPE and bound profiles.
- Both attained objective gaps below 10^{-4} from independent optima; cross-evaluating selected candidates differed by less than 10^{-10}.
- Python reported success after 29–40 iterations and 600–1,230 evaluations. R used its full 400-iteration cap in all three cases.
- One objective has different, equally good vegetation vectors; vector equality would misstate solution quality.

Bounded objective and arithmetic checks; no identical random path, general global guarantee or ecological accuracy.

Evidence: `reports/science/method-correspondence.md`; `docs/report/method-certification/optimizer-solution-quality/summary.csv`

One fixed LOVE example in both source engines

- R 4.6.0 / DISQOVER 0.9.14 and current Python read one shared `LOVEexample` input set. Hexadecimal readback verified every numeric input cell.
- Each engine evaluated 32 fixed candidates on eight active ring/repeat calls. All eight selected candidate indices agreed.
- Mean cover and pollen matched exactly on 40 finite cells each. Ring-factor and objective differences were at most 5.11×10^{-15} on 360 and eight finite cells, respectively.

This checks fixed-candidate source arithmetic, not search convergence, stochastic draws, timing or ecological accuracy.

Evidence: `docs/report/numerical-performance/love-source-comparison/report.json`

English LOVE: declared model profile

- 15 source sites, 1250–1750 BP, 30 targets; published PPE and fall speeds held fixed for retained taxa.
- Regional REVEALS plus ring-taxon-preserving LOVE; closed, nonnegative local cover.
- Uniform-start SLSQP: 300 iterations, 16,000 evaluations, objective tolerance 10^{-8}.
- 192/195 training fits complete; three failures also refused under a bounded 600-iteration check.

These are source-specific development choices, not universally optimal LRA parameters or the original R optimizer.

Evidence: `reports/science/english-lra-pollen-prediction.md`

English LOVE routes have different support

Route	Source axis and attempts	Retained result
Analytical	31 taxa; 15 folds	15/15 invalid compositions; all 54 local ring rows include negative cover.
Differential evolution	30 taxa; one pair, two requests	Both requests refused convergence; no held-out score.
Constrained SLSQP	30 taxa; 15 folds	14/15 folds predicted; BECKHAM had three training-fit refusals.

Excluding 159 Filipendula counts changed the taxon response. These development runs are not a paired solver ranking or a vegetation-accuracy comparison.

Evidence: `reports/science/english-lra-pollen-prediction.md`

Matched pollen comparison: result and limitation

	MAE		Hellinger	
	Direct	LRA	Direct	LRA
Training mean	.020496	.026273	.282082	.360671
Nearest	.019552	.021132	.255767	.286001
Inverse distance squared	.018579	.021088	.243673	.285415

14 paired English sites, 30 taxa; all three direct means remain lower after any one site is omitted. Prior exposure and unequal training information limit interpretation.

Evidence: `docs/report/pollen-reconstruction/author-lra-constrained/direct-pollen/paired-hellinger.json`

English basin radius: forward sensitivity only

- We replayed 42 frozen LRA vegetation estimates at 80%, 100% and 120% of each assumed target-lake radius.
- The actual forward consumer reproduced all 42 original pollen predictions exactly before target scoring.
- Inverse-distance-squared site-macro pollen MAE: .021032, .021088 and .021457 across those radii.
- On matched targets, direct pollen retains lower mean MAE and Hellinger for every method and radius factor.

14 prior-exposed English sites. Vegetation, regional cover and PPE were fixed; this does not calibrate basin geometry or admit Nordic LRA inputs.

Evidence: `docs/report/pollen-reconstruction/author-lra-constrained/radius-sensitivity/paired/report.json`; `reports/science/english-lra-pollen-prediction.md`

Model selection follows the target and support

- Fixed spatial baselines permit clear paired interpretation on observed pollen.
- A bounded RBF candidate improved one prior-exposed author comparison on common support but reduced coverage.
- Country-excluded tuning did not establish superiority over inverse-distance-squared in the Nordic development holdouts.
- A two-case synthetic PPE/transport winner reflects a transport-law mismatch and is not a real-core parameter recommendation.
- For LRA, input identity, chronology, basin geometry, PPE and denominator must be qualified before any parameter search is meaningful.

Evidence: `reports/science/model-use.md`; `reports/science/synthetic-model-selection/README.md`; `reports/needed/README.md`

HYDE and vegetation: retain the boundary

- Four-country HYDE source cropland grid sums match 84 country/year/scenario totals within 8.51×10^{-11} km^2.
- This checks modeled-source internal accounting, not pollen-model benefit.
- The retained release has conflicting rights and version declarations; formal CRS and historical calendar support remain unresolved.
- No independent vegetation measurement pair has been admitted for prediction accuracy.

Next experiments require source semantics and a matched ecological reference, with outcomes protected from model selection.

Evidence: `reports/science/hyde-cropland-source-account.md`; `reports/needed/README.md`

Why no vegetation-accuracy score exists yet

- EMPD2–NFI screen: 16 nearby candidates, zero measured pairs admitted for our benchmark.
- Färskesjön: the measured 7 m forest plot is 427 m from the lake pollen point; five volume classes do not supply 31-taxon landscape cover.
- Three Småland bogs: historical-map coverage and pollen anchors are published, but full numerical vegetation fractions are absent.
- A published southern-Swedish figure favors its authors' REVEALS estimates on nine rounded printed labels; it does not score our predictions or recover the full 26-taxon study.
- LandClim grid 779 names Bussjösjön and Bökesjön as input sites; its modeled cover is not an independent measured target for those pollen series.

A physical-site, time, taxon and landscape-support match is still required before ecological validation.

Evidence: `reports/science/vegetation-reference.md`; `docs/report/countries/sweden/pollen-model-lineage/report.json`; `docs/report/validation/empd2-nfi-pairing/report.json`

Open source decisions

- Abborrtjärnen: LandClim-selected age model, historical basin radius, eligible denominator, PPE use and rights.
- Bjäresjösjön: modern lake identity and nearby SEAD display intervals close; pollen-core identity and ecological event join remain refused.
- Solsø: 79 LandClim II samples versus 92 Neotoma source samples; the published age span retains 90, and no historical basin series is bound to the selected rows.
- Dalmutladdo (Norway): 72 source samples and 11 radiocarbon controls align; published 7.2 ha versus LandClim 8 ha, with no dated basin series. Zero LRA admission.
- Laihalampi (Finland): one LAIH core and 150 depths align; modern area claims agree conditionally, but selected ages and dated basin remain unqualified.
- Historical publication: exact source revisions and authenticated PFT/uncertainty transformation.
- Independent confirmation: new qualified cohort frozen before target inspection.

Evidence: `reports/needed/README.md`; `reports/science/solsoe-lra-input.md`; `reports/science/dalmutladdo-lra-input/README.md`; `reports/science/laihalampi-lra-input/README.md`